

Successive Differentiation

If $f(x)$ & its derivatives are all differentiable, we can find n th derivative of $f(x)$, i.e. $f^n(x)$

We denote n th derivative by

$$\frac{d^n f}{dx^n} \text{ or } D^n(f(x)) \text{ where } D^n = \frac{d^n}{dx^n}$$

n th derivatives of some standard functions—

1) $y = e^{ax}$

$$y_1 = ae^{ax}$$

$$y_2 = a^2 e^{ax}$$

Continuing in the same way

$$y_n = a^n e^{ax}$$

2) $y = (ax+b)^m$

$$y_1 = m(ax+b)^{m-1} (a)$$

$$y_2 = m(m-1)(ax+b)^{m-2} a^2$$

Continuing in the same way

$$y_n = m(m-1)\dots(m-(n-1))(ax+b)^{m-n} a^n$$

$$y_n = \frac{m!}{(m-n)!} a^n (ax+b)^{m-n} ; \text{ if } m > n \text{ (m +ve integer)}$$

$$y_n = 0 ; \text{ if } m < n$$

$$y_n = n! a^n ; \text{ if } m = n$$

Also if $m = -1$

$$y_n = (-1)(-2)\dots(-n)(ax+b)^{-1-n} a^n$$

$$= \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

3) $y = \log(ax+b)$

$$y_1 = \frac{0}{ax+b}$$

$$y_1 = a(ax+b)^{-1}$$

$$y_2 = a^2(-1)(ax+b)^{-2}$$

As done in the last case

$$y_n = (-1)^{n-1} \frac{(n-1)! a^n}{(ax+b)^n}$$

$$(4) \quad y = a^x$$

$$y_1 = a^x \log a$$

$$y_2 = a^x (\log a) \cdot \log a = a^x (\log a)^2$$

...

$$y_n = a^x (\log a)^n$$

$$(5) \quad y = \sin(ax+b)$$

$$y_1 = a \cos(ax+b) = a \sin(ax+b + \frac{\pi}{2})$$

$$y_2 = a^2 \cos(ax+b + \frac{\pi}{2}) = a^2 \sin(ax+b + 2\frac{\pi}{2})$$

...

$$y_n = a^n \sin(ax+b + n\frac{\pi}{2})$$

$$(6) \quad y = \cos(ax+b)$$

$$y_1 = -a \sin(ax+b) \cdot a$$

$$= a \cos(ax+b + \pi/2)$$

$$y_2 = -a^2 \sin(ax+b + \pi/2) = a^2 \cos(ax+b + 2\pi/2)$$

...

$$y_n = a^n \cos(ax+b + n\frac{\pi}{2})$$

$$y = e^{ax} \sin(bx+c)$$

$$y_1 = ae^{ax} \sin(bx+c) + e^{ax} \cos(bx+c) \cdot b$$

$$= e^{ax} [a \sin(bx+c) + b \cos(bx+c)]$$

$$\text{put } a = r \cos \alpha$$

$$b = r \sin \alpha$$

$$y_1 = e^{ax} [r \cos \alpha \sin(bx+c) + r \sin \alpha \cos(bx+c)]$$

$$= r e^{ax} [\sin(bx+c+\alpha)]$$

Again differentiating

$$y_2 = ae^{ax} [a \sin(bx+c) + b \cos(bx+c)] +$$

$$e^{ax} [ba \cos(bx+c) - b^2 \sin(bx+c)]$$

$$= a^2 e^{ax} \sin(bx+c) + 2ab \cos(bx+c) e^{ax} - b^2 e^{ax} \sin(bx+c)$$

$$= (a^2 - b^2) e^{ax} \sin(bx+c) + 2ab e^{ax} \cos(bx+c)$$

$$= r^2 (\cos^2 \alpha - \sin^2 \alpha) e^{ax} \sin(bx+c) + 2r^2 \cos \alpha \sin \alpha e^{ax} \cos(bx+c)$$

$$= r^2 \cos 2\alpha e^{ax} \sin(bx+c) + r^2 \sin 2\alpha e^{ax} \cos(bx+c)$$

$$= r^2 e^{ax} [\sin(bx+c+2\alpha)]$$

$$\vdots$$

$$y_n = r^n e^{ax} [\sin(bx+c+n\alpha)]$$

$$\text{II} y = e^{ax} \cos(bx+c)$$

$$y_n = e^{ax} r^n \cos(bx+c+n\alpha)$$

$$\text{where } r = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1} \frac{b}{a}$$

Find the n th derivative of

$$(1) \quad \frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

$$4x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

$$= A(x^2-1) + B(x+1) + C(x^2+1-2x)$$

$$A+C=0 \quad (1)$$

$$B-2C=4 \quad (2) \Rightarrow B=4+2C$$

$$-A+B+C=0 \quad (3) \Rightarrow -A+4+2C+C=0$$

$$-A+3C=-4 \quad (4)$$

Using (1) & (4)

$$\begin{array}{r} A+C=0 \\ -A+3C=-4 \\ \hline 4C=-4 \end{array}$$

$$C=-1$$

$$A=-C=1$$

$$B=A-C=1-(-1)=2.$$

$$y = \frac{4x}{(x-1)^2(x+1)} = \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} = (x-1)^{-1} + 2(x-1)^{-2} - (x+1)^{-1}$$

$$y_n = \frac{(-1)^n n! (1)^n}{(x-1)^{n+1}} + \left[2 \frac{d^n}{dx^n} (x-1)^{-2} \right] - \frac{(-1)^n n! (1)^n}{(x+1)^{n+1}}$$

$$\text{Now } \frac{d^n}{dx^n} (x-1)^{-2} = \frac{d^{n-1}}{dx^{n-1}} [(-2)(x-1)^{-3}]$$

$$= \frac{d^{n-2}}{dx^{n-2}} [(-2)(-3)(x-1)^{-4}]$$

$\vdots$

$$= (-2)(-3)\dots(-n+1)(x-1)^{-(n+2)}$$

$$= (-1)^n n! (n+1) / (x-1)^{n+2}$$

$$\therefore y_n = \frac{(-1)^n n!}{1} \left[\frac{1}{(x-1)^{n+1}} + \frac{2(n+1)}{(x-1)^{n+2}} - \frac{1}{(x+1)^{n+1}} \right]$$

eg: Find $\frac{d^2y}{dx^2}$ if $x = a(\theta + \sin \theta)$
 $y = a(1 - \cos \theta)$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{d\theta} \left(\frac{dy}{dx} \right) \frac{d\theta}{dx} \\ &= \frac{\cos \theta (1 + \cos \theta) - \sin \theta (-\sin \theta)}{(1 + \cos \theta)^2} \cdot \frac{1}{a(1 + \cos \theta)} \\ &= \frac{1}{a(1 + \cos \theta)^2} = \frac{1}{a(2 \cos^2 \frac{\theta}{2})^2} = \frac{1}{4a} \sec^4 \frac{\theta}{2} \end{aligned}$$

eg: If $y = e^{-kt} \cos(bt+c)$ then prove that
 $\frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + n^2 y = 0$ where $n^2 = k^2 + b^2$

Solu: $y_1 = -ke^{-kt} \cos(bt+c) + e^{-kt} b \sin(bt+c)$
 $= -e^{-kt} [k \cos(bt+c) + b \sin(bt+c)]$
 $y_2 = +ke^{-kt} [k \cos(bt+c) + b \sin(bt+c)] - e^{-kt} [-kb \sin(bt+c) + b^2 \cos(bt+c)]$

Now L.H.S : $y_2 + 2ky_1 + n^2 y =$
 $ke^{-kt} [k \cos(bt+c) + b \sin(bt+c)] - e^{-kt} [-kb \sin(bt+c) + b^2 \cos(bt+c)]$
 $+ 2k e^{-kt} [k \cos(bt+c) + b \sin(bt+c)]$
 $+ (b^2 + k^2) e^{-kt} \cos(bt+c)$
 $= (k^2 - b^2 - 2k^2 + b^2 + k^2) e^{-kt} \cos(bt+c) +$
 $(kb + kb - 2kb) e^{-kt} \sin(bt+c)$
 $= 0$

Q: $y = \tan^{-1} \left(\frac{1-x}{1+x} \right)$

$$\begin{aligned} y_1 &= \frac{1}{1 + \left(\frac{1-x}{1+x} \right)^2} \left[\frac{-1(1+x) - (1-x) \cdot 1}{(1+x)^2} \right] \\ &= \frac{(1+x)^2}{1+x^2+2x+1+x^2-2x} \times \frac{-2}{(1+x)^2} \\ &= \frac{-2}{2x^2+2} = \frac{-1}{1+x^2} = \frac{-1}{(x+i)(x-i)} \\ &= \frac{1}{2i} \left[\frac{1}{x+i} - \frac{1}{x-i} \right] \end{aligned}$$

Differentiating $(n-1)$ times w.r.t x

$$\begin{aligned} y_n &= \frac{1}{2i} \left[\frac{d^{n-1}}{dx^{n-1}} \frac{1}{x+i} - \frac{d^{n-1}}{dx^{n-1}} \frac{1}{x-i} \right] \\ &= \frac{1}{2i} \left[\frac{(-1)^{n-1}(n-1)!}{(x+i)^n} - \frac{(-1)^{n-1}(n-1)!}{(x-i)^n} \right] \end{aligned}$$

Now put $x = r \cos \theta$
 $1 = r \sin \theta$

$$\begin{aligned} &= \frac{1}{2i} \left[\frac{(-1)^{n-1}(n-1)!}{r^n (\cos \theta + i \sin \theta)^n} - \frac{(-1)^{n-1}(n-1)!}{r^n (\cos \theta - i \sin \theta)^n} \right] \\ &= \frac{1}{2i} (-1)^{n-1}(n-1)! \left[\frac{1}{r^n} (\cos n\theta - i \sin n\theta - \cos n\theta - i \sin n\theta) \right] \\ &= \frac{1}{2i} (-1)^{n-1}(n-1)! \frac{1}{r^n} (-2i \sin n\theta) \\ &= (-1)^n (n-1)! \frac{\sin n\theta}{r^n} \end{aligned}$$

But $r = \sqrt{x^2+1}$

$r^n = (x^2+1)^{n/2}$ OR $r = \frac{1}{\sin \theta} \Rightarrow r^n = \frac{1}{\sin^n \theta}$

$= (-1)^n (n-1)! \sin n\theta \sin^n \theta$ where $\theta = \tan^{-1} \left(\frac{1}{x} \right)$
 $= \cot^{-1} x.$

Leibnitz's theorem -

(4)

This thm provides us the n th derivative of product of two functions.

If $y = u.v$, where u & v are fns of x & their n th derivatives are known separately then

$$(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + u v_n.$$

Find the n th derivative of

① $y = e^x \log x = u.v$

$$\begin{aligned} y_n &= e^x \log x + {}^nC_1 e^x \frac{1}{x} + {}^nC_2 e^x \frac{(-1)}{x^2} + \dots + {}^nC_n e^x \frac{(-1)^{n-1} (n-1)!}{x^n} \\ &= e^x \log x + n \frac{e^x}{x} - \frac{n(n-1)}{2} \frac{e^x}{x^2} + \dots + (-1)^{n-1} (n-1)! \frac{e^x}{x^n} \end{aligned}$$

eg: If $y = \cos(m \sin^{-1} x)$ show that

$$(1-x^2) y_{n+2} = -(2n+1)x y_{n+1} + (m^2 - n^2) y_n = 0 \quad \& \text{ hence find}$$

$y_n(0)$

Solu :

$$y_1 = -\sin(m \sin^{-1} x) \cdot m \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} y_1 = -m \sin(m \sin^{-1} x)$$

Again differentiating

$$\frac{1}{2\sqrt{1-x^2}} \cdot (-2x) \cdot y_1 + \sqrt{1-x^2} y_2 = -m \cos(m \sin^{-1} x) \cdot m \frac{1}{\sqrt{1-x^2}}$$

$$-x y_1 + y_2 (1-x^2) = -m^2 y$$

By Leibnitz thm differ. n times

$$\begin{aligned} -x y_{n+1} - {}^nC_1 \cdot 1 y_n - {}^nC_2 \cdot 0 + (1-x^2) y_{n+2} - {}^nC_1 2x y_{n+1} - 2 y_n {}^nC_2 \\ = -m^2 y_n \end{aligned}$$

$$(1-x^2) y_{n+2} - (2n+1)x y_{n+1} + \left[-n - 2 \frac{(n^2-n)}{2} + m^2 \right] y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2-n^2)y_n = 0$$

Put $x=0$

$$y_{n+2} - (m^2-n^2)y_n = 0$$

For $n=0$

$$y_2 = (m^2-0^2)y_0 = m^2y_0$$

$$\text{For } n=1 \quad y_3 = (m^2-1)y_1$$

$$\text{For } n=2 \quad y_4 = (m^2-4)y_2 = (m^2-4)m^2y_0$$

$$\text{For } n=3 \quad y_5 = (m^2-9)y_3 = (m^2-9)(m^2-1)y_1$$

...

$$y_n(0) = \begin{cases} m^2(m^2-4)(m^2-16)\dots[m^2-(n-2)^2]y_0 & n \text{ ~~odd~~ even} \\ (m^2-1)(m^2-9)(m^2-25)\dots[m^2-(n-2)^2]y_1 & n \text{ ~~even~~ odd} \end{cases}$$

$$\text{But } y_0(0) = 1$$

$$y_1(0) = 0$$

$$y_n(0) = 0 \quad n \text{ odd}$$

$$= m^2(m^2-4)\dots(m^2-(n-2)^2) \quad n \text{ even.}$$

eg: $y = a \cos(\log x) + b \sin(\log x)$ s.t

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

solu: $y_1 = -a \sin(\log x) \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x}$

$$xy_1 = -a \sin(\log x) + b \cos(\log x)$$

$$xy_2 + y_1 = -a \cos(\log x) \frac{1}{x} - b \cos(\log x) \cdot \frac{1}{x}$$

$$x^2 y_2 + xy_1 = -y$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2-n^2)y_n = 0$$

Put $x=0$

$$y_{n+2} - (m^2-n^2)y_n = 0$$

For $n=0$

$$y_2 = (m^2-0^2)y_0 = m^2y_0$$

$$\text{For } n=1 \quad y_3 = (m^2-1)y_1$$

$$\text{For } n=2 \quad y_4 = (m^2-4)y_2 = (m^2-4)m^2y_0$$

$$\text{For } n=3 \quad y_5 = (m^2-9)y_3 = (m^2-9)(m^2-1)y_1$$

...

$$y_n(0) = \begin{cases} m^2(m^2-4)(m^2-16)\dots[m^2-(n-2)^2]y_0 & n \text{ ~~odd~~ even} \\ (m^2-1)(m^2-9)(m^2-25)\dots[m^2-(n-2)^2]y_1 & n \text{ ~~even~~ odd} \end{cases}$$

$$\text{But } y_0(0) = 1$$

$$y_1(0) = 0$$

$$y_n(0) = 0 \quad n \text{ odd}$$

$$= m^2(m^2-4)\dots(m^2-(n-2)^2) \quad n \text{ even.}$$

eg: $y = a \cos(\log x) + b \sin(\log x)$ s.t

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

solu: $y_1 = -a \sin(\log x) \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x}$

$$xy_1 = -a \sin(\log x) + b \cos(\log x)$$

$$xy_2 + y_1 = -a \cos(\log x) \frac{1}{x} - b \cos(\log x) \cdot \frac{1}{x}$$

$$x^2 y_2 + xy_1 = -y$$

Differentiating n times by Leibnitz thm

(5)

$$x^2 y_{n+2} + nC_1 2xy_{n+1} + nC_2 2y_{n+0} + xy_{n+1} + 1nC_1 \cdot 1 \cdot y_n = -y_n$$

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

eg: If $y = (\sin^{-1}x)^2$ p.T

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0. \text{ \& hence find } y_n(0)$$

Solu:

$$y_1 = 2\sin^{-1}x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$y_2 = 2 \cdot \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-x^2}} + 2\sin^{-1}x \cdot \frac{1}{(1-x^2)^{3/2}} \left(-\frac{1}{2}\right)(-2x)$$

$$= \frac{2}{1-x^2} + \frac{2x}{(1-x^2)^{3/2}} \sin^{-1}x.$$

$$(1-x^2)y_2 = 2 + 2xy_1$$

Differentiating n times

$$(1-x^2)y_{n+2} + nC_1(-2x)y_{n+1} + nC_2(-2)y_n = 0 + xy_{n+1} + nC_1 \cdot 1 \cdot y_n$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - 2 \frac{n(n-1)}{2} y_n = xy_{n+1} + ny_n$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0.$$

For $x=0$ eq takes the form

$$y_{n+2} = n^2 y_n \quad n > 0$$

$$y_2 = 2 ; y_3 = 1^2 y_1 ; y_4 = 2^2 y_2 = 2^2 \cdot 2 ; y_5 = 3^2 y_3 = 3^2 \cdot 1^2 y_1$$

$$y_6 = 4^2 y_4 = 4^2 \cdot 2^2 \cdot 2 ; y_7 = 5^2 y_5 = 5^2 \cdot 3^2 \cdot 1^2 y_1 \text{ \& so on}$$

$$\therefore y_n = (n-2)^2 (n-4)^2 \dots 4^2 \cdot 2^2 \cdot 2 \text{ \& even}$$

$$= 1^2 \cdot 3^2 \cdot 5^2 \dots (n-2)^2 y_1 \text{ \& n odd}$$

But $y_1(0) = 0 \quad \therefore$

$$y_n(0) = \begin{cases} 0 & n \text{ odd} \\ (n-2)^2(n-4)^2 \dots 4^2 \cdot 2^2 \cdot 2 & n \text{ even} \end{cases}$$

eg: If $x = \sin at$; $y = \cos at$, show that

$$(1-x^2)y_1^2 = a^2(1-y_2) \text{ \& hence deduce that}$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-a^2)y_n = 0$$

Solu :

$$y_1 = \frac{dy/dt}{dx/dt} = -\frac{a \sin at}{a \cos at} \quad \boxed{-\tan at}$$

$$= \frac{-a(\sqrt{1-\cos^2 \alpha t})}{\sqrt{1-\sin^2 \alpha t}} = \frac{-a\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

$$y_1^2 = \frac{(1-y^2)a^2}{1-x^2}$$

$$(1-x^2) y_1^2 = a^2(1-y^2)$$

Differentiating we get

$$2(1-x^2)y_1 y_2 + \frac{d}{dx}(1-x^2) \cdot y_1^2 = a^2 [0 - 2y y_1]$$

$$x(1-x^2)y_2 - x^2y_1 = -x^2y$$

$$(1-x^2)y_2 - xy_1 = -a^2y$$

Differentiating n times

$$(1-x^2)y_{n+2} - 2x^n C_1 y_{n+1} - 2^n C_2 y_n - x y_{n+1} - {}^n C_1 \cdot y_n = -a^2 y_n$$

$$(1-x^2)y_{n+2} - 2nx y_{n+1} - \frac{2(n-1)n}{2} y_n - x y_{n+1} - n y_n + a^2 y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-a^2)y_n = 0$$

$$r^{(k)} = \mathbf{1} - \mathbf{R}$$

Taylor's Expansion -

Let f be a fn. defined on $[a, b]$ such that

- (i) $f^{(n-1)}$ is continuous on $[a, b]$
- (ii) $f^{(n-1)}$ is derivable on $]a, b[$

then there exists a real number $c \in]a, b[$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(b-a)^n}{n!}f^{(n)}(c) \quad \text{---(A)}$$

Suppose x is any point b/w a & b . If f satisfies all the conditions of Taylor's thm on $[a, b]$, then it also satisfies the same conditions on $[a, x]$. $\therefore$ we will have

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(x-a)^n}{n!}f^{(n)}(c). \quad \text{---(B)}$$

where c lies b/w a & x .

The last term $\frac{(x-a)^n}{n!}f^{(n)}(c)$ gives the error in taking the polynomial as an approximation of $f(x)$. This is called the "remainder after n terms" & is usually denoted by R_n .

Now if we are given a fn $f(x)$ we can write its Taylor's series using Taylor's expansion (obviously only if the conditions are satisfied). The series will converge iff the limit of remainder term exist i.e. $\lim_{n \rightarrow \infty} R_n$ exist.

Also the series will converge to $f(x)$ i.e. the sum of the series will be $f(x)$ iff $\lim_{n \rightarrow \infty} R_n = 0$.

Proof: Let us first observe that condition (i) in the Statement of the theorem implies that $f, f', \dots, f^{n-1}$ are all defined and continuous on $[a, b]$.

Consider the fn $F(x)$ defined on $[a, b]$ by setting

$$F(x) = f(b) - f(x) - (b-x)f'(x) - \frac{(b-x)^2}{2!}f''(x) - \dots - \frac{(b-x)^{n-1}}{(n-1)!}f^{(n-1)}(x) - A(b-x)^n.$$

A is a suitably chosen constant.

Now

- (1) Since $f, f', \dots, f^{n-1}$ are all cts on $[a, b]$ & since $x \rightarrow (b-x)^k$ is cts fn on $[a, b]$ for each +ve integer k , $\therefore F$ is cts on $[a, b]$
- (2) Since $f, f', \dots, f^{n-1}$ are all derivable on $]a, b[$ & since $x \rightarrow (b-x)^k$ is derivable on $]a, b[$ $\forall$ +ve integer k , therefore F is derivable on $]a, b[$.

- (3) Let us choose A so that $F(a) = F(b)$. Since $F(b) = 0 \therefore F(a) = 0$ i.e.

$$f(b) - f(a) - (b-a)f'(a) - \frac{(b-a)^2}{2!}f''(a) - \dots - \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) - A(b-a)^n = 0 \quad \text{---(i)}$$

Now (1), (2), (3) are conditions of Rolle's thm $\therefore \exists$ a real no. $c \in]a, b[$ s.t. $F'(c) = 0$

Since

$$F'(x) = -f'(x) - (b-x)f''(x) + f'(x) + (b-x)f''(x) - \frac{(b-x)^2}{2!}f'''(x) - \dots - \frac{(b-x)^{n-1}}{(n-1)!}f^{(n)}(x) + nA(b-x)^{n-1}$$

$$\therefore -\frac{(b-c)^{n-1}}{(n-1)!}f^{(n)}(c) + nA(b-c)^{n-1} = 0$$

$$\therefore A = \frac{f^{(n)}(c)}{(n-1)!n} = \frac{f^{(n)}(c)}{n!} \quad \text{---(ii)}$$

From (i) & (ii)

$$f(b) = f(a) + (b-a)f'(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{(b-a)^n}{n!} f^n(c).$$

Maclaurin's Series -

If in above Taylor's series, we put $a=0$, we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + \dots$$

This is called Maclaurin series of $f(x)$.

eg:

$$f(x) = e^x \quad \forall x \in \mathbb{R}.$$

$$f^n(x) = e^x \quad \forall n \in \mathbb{N} \quad \forall x \in \mathbb{R}$$

Lagrange's remainder after n terms

$$R_n(x) = \frac{x^n}{n!} f^n(c) = \frac{x^n}{n!} e^c.$$

Now we will show that $\lim_{n \rightarrow \infty} R_n(x) = 0$

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} e^c = 0$$

Since $c \in]0, x[\therefore$

$$0 < c < x$$

$$e^0 < e^c < e^x$$

$$1 < e^c < e^x$$

$\therefore e^c$ is bdd & hence has finite value

$$\text{Now } \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad \therefore \lim_{n \rightarrow \infty} R_n(x) = 0$$

Now M.S will be

$$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

eg: Use Maclaurin's series to expand $\tan x$ upto x^5

$$f(x) = \tan x$$

$$f(0) = 0$$

$$f'(x) = \sec^2 x = 1 + \tan^2 x$$

$$f'(0) = 1$$

$$f''(x) = 0 + 2 \tan x \sec^2 x = 2 \tan x (1 + \tan^2 x) \quad f''(0) = 0$$

$$f'''(x) = 2 \sec^2 x + 2 \cdot 3 \tan^2 x \sec^2 x \quad f'''(0) = 2$$

$$= 2 \sec^2 x + 6 \tan^2 x \sec^2 x$$

$$= 2(1 + \tan^2 x) + 6 \tan^2 x (1 + \tan^2 x)$$

$$= 2 + 8 \tan^2 x + 6 \tan^4 x$$

$$f^{(4)}(x) = 0 + 16 \tan x \sec^2 x + 24 \tan^3 x \sec^2 x$$

$$= 16 \tan x (1 + \tan^2 x) + 24 \tan^3 x (1 + \tan^2 x)$$

$$= 16 \tan x + 40 \tan^3 x + 24 \tan^5 x \quad f^{(4)}(0) = 0$$

$$f^{(5)}(x) = 16 \sec^2 x + 40 \cdot 3 \tan^2 x \sec^2 x + 120 \tan^4 x \sec^2 x$$

$$= 16 \sec^2 x + 120 \tan^2 x \sec^2 x + 120 \tan^4 x \sec^2 x$$

$$f^{(5)}(0) = 16$$

& so on

$$\tan x = f(0) + \frac{x^1}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0)$$

$$+ \frac{x^5}{5!} f^{(5)}(0) + \dots$$

$$= 0 + x \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} \cdot 2 + \frac{x^4}{4!} \cdot 0 + \frac{x^5}{5!} \cdot 16$$

$$= x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

eg: Expand $e^{\sin x}$ by Maclaurin's series or otherwise upto the term containing x^4 .

$$\begin{aligned}
 e^{\sin x} &= 1 + \sin x + \frac{(\sin x)^2}{2!} + \frac{(\sin x)^3}{3!} + \dots \\
 &= 1 + \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right) + \frac{1}{2!} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)^2 \\
 &\quad + \frac{1}{3!} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)^3 + \dots \\
 &= 1 + \left(x - \frac{x^3}{6} + \dots\right) + \left(\frac{x^2}{2} - \frac{2x^4}{2!3!} + \dots\right) + \frac{1}{3!} (x^3 - \dots) \\
 &\quad + \frac{1}{4!} (x^4 - \dots) \\
 &= 1 + x - \frac{x^3}{6} + \frac{x^2}{2} - \frac{x^4}{6} + \frac{x^2}{6} + \frac{x^4}{24} + \dots \\
 &= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{eg: } f(x) = e^x \cos x &= \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) \\
 &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + x - \frac{x^3}{2!} + \frac{x^5}{4!} - \dots +
 \end{aligned}$$

$$\begin{aligned}
 &\quad \frac{x^4}{2!} - \frac{x^4}{2!2!} + \frac{x^5}{2!4!} + \dots + \frac{x^3}{3!} - \frac{x^5}{3!2!} + \frac{x^7}{3!4!} + \dots \\
 &\quad + \frac{x^4}{4!} - \dots \\
 &= 1 + x - \frac{x^3}{3} - \frac{1}{6}x^4 \\
 &= 1 + x - \frac{2x^3}{3!} - \frac{2^2 x^4}{4!} - \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{eg: } \log(1+e^x) &= \log\left[1 + \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\right)\right] \\
 &= \log\left[2 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots\right] \\
 &= \log 2 \left[1 + \left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + \frac{x^4}{48} + \dots\right)\right] \\
 &= \log 2 + \log\left[1 + \left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + \frac{x^4}{48} + \dots\right)\right] \\
 &= \log 2 + \left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + \frac{x^4}{48} + \dots\right) - \frac{1}{2}\left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + \frac{x^4}{48} + \dots\right)^2 \\
 &\quad + \frac{1}{3}\left(\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{12} + \dots\right)^3 - \frac{1}{4}\left(\frac{x}{2} + \dots\right)^4 \\
 &= \log 2 + \left(\frac{x}{2}\right) + \left(\frac{x^2}{4} - \frac{x^2}{8}\right) + \left(\frac{x^3}{12} - \frac{1}{2} \cdot \frac{2x^3}{8} + \frac{1}{3} \cdot \frac{x^3}{8}\right) + \dots \\
 &\quad + \left(\frac{x^4}{48} - \frac{1}{2} \cdot \frac{x^4}{16}\right) + \left(\frac{1}{48}x^4 - \frac{1}{64}x^4\right) \\
 &= \log 2 + \frac{x}{2} + \frac{x^2}{8} - 0x^3 + \frac{2x^4}{192} + \dots
 \end{aligned}$$

eg: Expand $\sin x$ in powers of $(x - \pi/2)$. ; $f(x) = \sin x$; $f(\pi/2) = 1$

$$f'(x) = \cos x$$

$$f'(\pi/2) = 0$$

$$f''(x) = -\sin x$$

$$f''(\pi/2) = -1$$

$$f'''(x) = -\cos x$$

$$f'''(\pi/2) = -0$$

$$f^{(iv)}(x) = +\sin x$$

$$f^{(iv)}(\pi/2) = 1$$

$$f^{(v)}(x) = \cos x$$

$$f^{(v)}(\pi/2) = 0 \quad \& \text{ so on}$$

$$\begin{aligned}
 \sin x &= f\left(\frac{\pi}{2}\right) + (x - \pi/2)f'(\pi/2) + \frac{(x - \pi/2)^2}{2!}f''(\pi/2) + \frac{(x - \pi/2)^3}{3!}f'''(\pi/2) \\
 &\quad + \frac{(x - \pi/2)^4}{4!}f^{(iv)}(\pi/2) + \dots
 \end{aligned}$$

$$\begin{aligned}
&= 1 + (x - \pi/2) \cdot 0 + \frac{(x - \pi/2)^2}{2!} (-1) + \frac{(x - \pi/2)^3}{3!} \cdot 0 + \\
&\quad \frac{(x - \pi/2)^4}{4!} \cdot 1 + \frac{(x - \pi/2)^5}{5!} \cdot 0 + \dots \\
&= 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \dots
\end{aligned}$$

~~Maclaurin's~~ Approximation.

Maclaurin's Series of some standard fns-

① $f(x) = \sin x$
 $f^n(x) = \sin(x + n\pi/2)$
 $f^n(0) = \sin(n\pi/2)$

$$\begin{aligned}
\therefore \sin x &= f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots \\
&= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} - \dots \\
&= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots
\end{aligned}$$

② $f(x) = \cos x$
 $f^n(0) = \cos(n\pi/2)$

$$\begin{aligned}
f(x) = \cos x &= 1 + x \cdot 0 + \frac{x^2}{2!} (-1) + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \cdot 1 + \dots \\
&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots
\end{aligned}$$

③ $f(x) = e^x$ (already done)

④ $\log(1+x) = f(x)$

$$f^n(x) = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n} \quad \therefore f^n(0) = \frac{(-1)^{n-1} (n-1)!}{[1]^n}$$

$$\begin{aligned}\log(1+x) &= \log(1+0) + x \cdot 0! + \frac{x^2}{2!} (-1)^1 1! + \frac{x^3}{3!} (-1)^2 2! + \\ &\quad \frac{x^4}{4!} (-1)^3 3! + \dots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots\end{aligned}$$

⑤ $f(x) = \tan x$ (already done)

eg: Apply Taylor's theorem to estimate the value of $f(11/10)$ where

$$f(x) = x^3 + 3x^2 + 15x - 10$$

$$f'(x) = 3x^2 + 6x + 15$$

$$f''(x) = 6x + 6$$

$$f'''(x) = 6$$

We apply Taylor's thm for $f(x)$ about $x = \frac{11}{10}$ & $a = 1$

~~$$f(x) = f\left(-\frac{1}{10}\right) + \left(x + \frac{1}{10}\right) f'\left(-\frac{1}{10}\right) + \left(x + \frac{1}{10}\right)^2 f''\left(-\frac{1}{10}\right) \frac{1}{2!} + \left(x + \frac{1}{10}\right)^3 f'''\left(-\frac{1}{10}\right) \frac{1}{3!}$$~~

$$f(x) = f\left(\frac{11}{10}\right) = f(1) + \left(\frac{11}{10} - 1\right) f'(1) + \left(\frac{11}{10} - 1\right)^2 \frac{1}{2!} f''(1) +$$

$$\frac{1}{3!} \left(\frac{11}{10} - 1\right)^3 f'''(1) + \dots$$

$$= 9 + \frac{1}{10} \cdot 24 + \frac{1}{100} \cdot \frac{12}{2!} + \frac{1}{3!} \cdot \frac{1}{1000} \cdot 6$$

$$= 9 + 2.4 + 0.06 + 0.001$$

$$= 11.461$$

Error & Approximation

Another form of Taylor's thm

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

$a=x \quad b=x+h$

Let $y = f(x)$. we give an increment δx in x . The corresponding increment in y will be δy

$$y + \delta y = f(x + \delta x)$$

$$\delta y = f(x + \delta x) - f(x)$$

$$= \left[f(x) + \delta x f'(x) + \frac{\delta x^2}{2!} f''(x) + \frac{\delta x^3}{3!} f'''(x) + \dots \right] - f(x)$$

$$= \delta x f'(x) + \frac{\delta x^2}{2!} f''(x) + \frac{\delta x^3}{3!} f'''(x) + \dots$$

Since δx is small $\therefore$ neglecting higher powers of δx

$$\delta y = \delta x f'(x) = \frac{dy}{dx} \delta x \quad \text{approx.}$$

If δx is error in x then $\frac{\delta x}{x}$ is called relative error & $\frac{\delta x}{x} \times 100$ is called percentage error.

Q: Find the change in the total surface area of a right circular cone when

(i) The radius is constant & the altitude changes by a small amount δh .

(ii) the altitude is constant & the radius changes by a small radius δr .

Solo: Surface area $S = \pi r^2 + \pi r l$
 $= \pi r^2 + \pi r \sqrt{r^2 + h^2}$



(i) altitude changes by δh

$$\frac{dS}{dh} = 0 + \frac{\pi r \cdot 2h}{2\sqrt{r^2 + h^2}} = \frac{\pi r h}{\sqrt{r^2 + h^2}}$$

$$\therefore \delta S = \frac{dS}{dh} \delta h = \frac{\pi r h}{\sqrt{r^2 + h^2}} \delta h$$

(ii) radius changes by δr

$$\begin{aligned} \frac{dS}{dr} &= 2\pi r + \pi \frac{1}{2} \frac{r \cdot 2r}{\sqrt{h^2 + r^2}} + \pi \sqrt{r^2 + h^2} \\ &= 2\pi r + \frac{\pi (r^2 + r^2 + h^2)}{\sqrt{r^2 + h^2}} \\ &= 2\pi r + \frac{\pi (2r^2 + h^2)}{\sqrt{r^2 + h^2}} \end{aligned}$$

$$\therefore \delta S = \frac{dS}{dr} \delta r = \left[2\pi r + \frac{\pi (2r^2 + h^2)}{\sqrt{r^2 + h^2}} \right] \delta r$$

eg: A soap bubble of radius 2 cm shrinks to radius 1.9 cm. Estimate the decrease in (i) Volume (ii) surface area

Solo: (i) $V = \frac{4}{3} \pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2$

$$\delta V = \frac{dV}{dr} \delta r = 4\pi r^2 \Big|_{r=2} (-0.1) = 50.24 (-0.1) = -5.024 \text{ cm}^3$$

(ii) $S = 4\pi r^2$

$$\frac{dS}{dr} = 8\pi r \Big|_{r=2} (-0.1) = -5.024 \text{ cm}^2$$

* Expand $\tan^{-1} x$ in powers of $(x-1)$ A: $\frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 - \dots$

* Expand poly. $2x^3 + 7x^2 + x - 1$ in powers of $(x-2)$

A: $45 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$

Infinite Series

①

Significance - Suppose a system (physical) can be represented by a series say an infinite series. If the series is convergent that will mean the system cgo to some finite number & hence the system can be summed up. Using this property of summation of series we can easily study the system & its properties. Now in case series is dgt then (the system will expand to ∞) we can't be able to study the system. This is the significance of studying infinite series.

Sequence - Its a fn from the set N of natural numbers to the set R of real numbers. represented by $\langle \rangle$

$$\langle \rangle : N \rightarrow R$$

$$\langle 1, 2, 3, \dots \rangle = \langle a_1, a_2, a_3, \dots \rangle$$

where $a_i \in R$

We can also call the sequence as an ordered set of numbers $a_1, a_2, \dots$. If the no. of terms are infinite then we call it infinite sequence otherwise finite sequence. The terms $a_1, a_2, \dots$ are called 1st term, 11nd term etc.

eg $\langle n^2 \rangle = \langle 1, 4, 9, 16, \dots \rangle$

n^2 represent n th term of the sequence

$$\langle \frac{1}{n} \rangle = \langle 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \rangle$$

Convergent Sequence - A seq $\langle a_n \rangle$ is said to converge to a number l if for given $\epsilon > 0$, $\exists$ a +ve integer m (depending upon ϵ) such that

$$|a_n - l| < \epsilon \quad \forall n > m$$

If the sequence converges to l then l is called the limit of the sequence. If limit exists its always unique.

If a sequence does not converge then it may be divergent or oscillatory. (Divergent means convergence towards $\pm\infty$). & oscillatory if neither converge nor diverge.

egs $\langle \frac{1}{n} \rangle = \langle 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots \rangle \rightarrow 0$

$$\langle n \rangle = \langle 1, 2, 3, \dots, n, \dots \rangle \rightarrow +\infty$$

$$\langle -n \rangle = \langle -1, -2, -3, \dots, -n, \dots \rangle \rightarrow -\infty$$

$$\langle (-1)^n \rangle = \langle -1, 1, -1, 1, \dots \rangle \text{ oscillatory sequence.}$$

Thm: Suppose $\langle a_n \rangle \rightarrow a$; $\langle b_n \rangle \rightarrow b$ then

$$\langle a_n \pm b_n \rangle \rightarrow a \pm b$$

$$\langle a_n b_n \rangle \rightarrow ab$$

$$\langle a_n / b_n \rangle \rightarrow a/b \text{ provided } b \neq 0$$

Bounded & Monotonic seq:

Let $\langle a_n \rangle$ is such that $\exists K$ which satisfies

$K \leq a_n \quad \forall n$ then $\langle a_n \rangle$ is called bounded below seq & K is called lower bound.

If $\exists K$ s.t. $a_n \leq K \quad \forall n$ then $\langle a_n \rangle$ is called bounded above seq. & K is called upper bound.

If $\exists$ both K & K simultaneously s.t.

$K \leq a_n \leq K \quad \forall n$ the seq is called a bdd seq.

Now suppose $\langle a_n \rangle$ is such that $a_n \leq a_{n+1} \quad \forall n$ then $\langle a_n \rangle$ is called monotonically increasing seq.

② $\langle a_n \rangle$ satisfies $a_{n+1} \leq a_n \forall n$ then $\langle a_n \rangle$ is called monotonically decreasing seq.

Results: A monotonically increasing seq which is bdd above is always cgt.

A monotonically dec. seq which is bdd below is always cgt.

A monotonically (inc. or dec) seq which is bdd is always cgt.

Cauchy's Thm: The seq $\langle a_n \rangle$ of real numbers cgs iff $\forall \epsilon > 0, \exists N$ such that $|a_n - a_m| < \epsilon \forall n, m > N$.

This thm has one advantage that we need not know the limit of seq to prove its convergence.

Thm: Convergence of a sequence implies boundedness but not converse.

eg: $\langle (-1)^n \rangle = \langle -1, 1, -1, 1, \dots \rangle$ is bdd by 1 & -1 but its not a convergent sequence.

Infinite series -

An expression of the form $a_1 + a_2 + a_3 + \dots = \sum a_n$ is called an infinite series.

$$\text{Let } S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$\vdots$$

$$S_n = a_1 + \dots + a_n$$

Consider the seq $\langle S_1, S_2, \dots \rangle$ of the sums S_i 's. This sequence, called the seq. of partial sums. Now if the seq $\langle S_n \rangle$ converges $\sum a_n$ cgs if $\langle S_n \rangle$ dgs then $\sum a_n$ dgs (same for oscillatory)

If the series is cgt then the limit of sequence of S_n is called the sum of the series.

Addition or omission of finite no. of terms from an infinite series does not change its nature of cgs, dgs or oscillation.

eg: The series $1+2+3+\dots$ dgs.

$$S_n = 1+2+3+\dots+n$$

$$= \frac{1}{2} n(n+1)$$

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2} n(n+1) = \infty \quad (\text{limit of seq method}).$$

or Since limit of seq is $\infty \therefore$ seq $\nrightarrow$ hence series dgs

eg: The series $2-2+2-2+2-\dots$ oscillates

$$S_n = \begin{matrix} 2 & n \text{ odd} \\ 0 & n \text{ even} \end{matrix}$$

$\therefore S_n = \langle 2, 0, 2, 0, \dots \rangle$ which is oscillatory seq $\nrightarrow$ hence series oscillates.

Geometric Series - show that $a + ar + ar^2 + \dots + ar^{n-1} + \dots$ is cgt for $|r| < 1$ & dgt if $|r| > 1$ & oscillates if $r \leq -1$

Consider the seq $\langle S_1, S_2, \dots \rangle$ of the sums S_i 's. This sequence, called the seq. of partial sums. Now if the seq $\langle S_n \rangle$ converges $\sum a_n$ cgs if $\langle S_n \rangle$ dgs then $\sum a_n$ dgs (same for oscillatory)

If the series is cgt then the limit of sequence of S_n is called the sum of the series.

Addition or omission of finite no. of terms from an infinite series does not change its nature of cgs, dgs or oscillation.

eg: The series $1+2+3+\dots$ dgs.

$$S_n = 1+2+3+\dots+n$$

$$= \frac{1}{2}n(n+1)$$

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2}n(n+1) = \infty \quad (\text{limit of seq method}).$$

or Since limit of seq is $\infty \therefore$ seq $\nrightarrow$ hence series dgs

eg: The series $2-2+2-2+2-\dots$ oscillates

$$S_n = \begin{matrix} 2 & n \text{ odd} \\ 0 & n \text{ even} \end{matrix}$$

$\therefore S_n = \langle 2, 0, 2, 0, \dots \rangle$ which is oscillatory seq $\nrightarrow$ hence series oscillates.

Geometric Series - show that $a+ar+ar^2+\dots+ar^{n-1}+\dots$ is cgt for $|r| < 1$ & dgt $|r| > 1$ & oscillates if $r \leq -1$

Case ① $|x| < 1$

$$S_n = \frac{a(1-x^n)}{1-x}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a}{1-x} - \frac{a}{1-x} \lim_{n \rightarrow \infty} x^n$$

$$= \frac{a}{1-x} - \frac{a}{1-x} \times 0$$

$$\because |x| < 1 \Rightarrow \lim_{n \rightarrow \infty} x^n = 0$$

$$= \frac{a}{1-x} \text{ finite quantity}$$

Case ②: $|x| \geq 1$

If $x=1$ then

$$S_n = a + a + \dots + a \text{ (n times)}$$

$$= na$$

$$\lim_{n \rightarrow \infty} S_n = \infty$$

If $x > 1$ then

$$S_n = \frac{a(1-x^n)}{1-x}$$

$$\text{where } \lim_{n \rightarrow \infty} x^n = \infty$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \infty$$

Case ③: $x \leq -1$

If $x = -1$ then

$$S_n = a - a + a - a - \dots + (-1)^{n-1} a = 0 \quad ; \quad \text{even}$$

$$a \quad ; \quad \text{odd oscillate}$$

If $x < -1$ then $x = -p$ then

$$x^n = (-1)^n p^n$$

$$S_n = \frac{a(1-x^n)}{1-x} = \frac{a(1-(-1)^n p^n)}{1+p}$$

$\lim_{n \rightarrow \infty} S_n \rightarrow -\infty$ or ∞ as n is even or odd $\therefore$ oscillatory.

Positive term Series - An infinite series in which from & after certain term, all the terms are positive is called positive term Series eg

$1 - 2 + 3 - 4 + 5 + 6 + 7 + 8 + \dots$ is a positive term Series.

If all terms of a series are +ve then the seq of p.s will be a monotonically increasing sequence & if it's bdd also then the seq of p.s & hence the series will be convergent.

Necessary condition for convergence of a series -

If $\sum a_n$ is cgt then $\lim_{n \rightarrow \infty} a_n = 0$

Proof: Consider $S_n = a_1 + a_2 + \dots + a_n$

$$S_{n-1} = a_1 + a_2 + \dots + a_{n-1}$$

Now since $\sum a_n$ is cgt $\therefore \lim_{n \rightarrow \infty} S_n$ will be finite say 's'

$$\text{then } \lim_{n \rightarrow \infty} S_n = s$$

$$\text{Also } \lim_{n \rightarrow \infty} S_{n-1} = s$$

$$\therefore \lim_{n \rightarrow \infty} (S_n - S_{n-1}) = \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} = s - s = 0$$

$$\lim_{n \rightarrow \infty} a_n = 0$$

This is necessary but not sufficient condition for cgs.

eg $\sum_{n=1}^{\infty} \frac{1}{n}$ In this case $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

but this is a dgt series.

We can use this to check the divergence of a series. If

$\lim_{n \rightarrow \infty} a_n \neq 0$ then series will be dgt series.

$$\sum_{n=1}^{\infty} \frac{1}{n+1} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n}} = 1 \neq 0$$

& therefore the series is dgt.

eg: $\sum \sqrt{\frac{n}{2(n+1)}}$ Ans: dgt. $\because \lim_{n \rightarrow \infty} a_n = \frac{1}{\sqrt{2}}$

Tests for Cgs & Dgs of an Infinite Series—

Since its not easy to find n th partial sum of the series $\therefore$ we can't always use $\lim_{n \rightarrow \infty} S_n$ method to check cgs or dgs. So we use different tests for this.

① Comparison test—Let $\sum u_n$ & $\sum v_n$ are two series.

(a) If $u_n \leq v_n \forall n$ & $\sum v_n$ is cgt then $\sum u_n$ also cgt

(b) If $u_n \leq v_n \forall n$ & $\sum u_n$ is dgt then $\sum v_n$ also dgt

(c) If $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = L \neq 0$ & finite, then $\sum u_n$ & $\sum v_n$ cgs or dgs together.

② Harmonic Series or p -series.

The infinite series $\sum \frac{1}{n^p}$ is known as harmonic series or p -series. This series is cgt for $p > 1$ & dgt for $p \leq 1$.

Proof: We have the series

$$\sum \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots$$

Case (1): $p > 1$

$$\begin{aligned} \sum \frac{1}{n^p} &= \frac{1}{1^p} + \left(\frac{1}{2^p} + \frac{1}{3^p} \right) + \left(\frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} \right) + \left(\frac{1}{8^p} + \dots + \frac{1}{15^p} \right) + \dots \\ &< \frac{1}{1^p} + \left(\frac{1}{2^p} + \frac{1}{2^p} \right) + \left(\frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p} \right) + \left(\frac{1}{8^p} + \dots + \frac{1}{8^p} \right) + \dots \\ &= \frac{1}{1^p} + \frac{2}{2^p} + \frac{4}{4^p} + \frac{8}{8^p} + \dots \end{aligned}$$

$$\sum \frac{1}{n^p} < \frac{1}{1^p} + \frac{1}{2^{p-1}} + \frac{1}{4^{p-1}} + \frac{1}{8^{p-1}} + \dots$$

(G.S with CR $\frac{1}{2^{p-1}}$ low for

$$= 1 + \frac{1}{2^{p-1}} + \frac{1}{(2^2)^{p-1}} + \frac{1}{(2^3)^{p-1}} + \dots$$

$$= 1 + \frac{1}{2^{p-1}} + \frac{1}{(2^{p-1})^2} + \frac{1}{(2^{p-1})^3} + \dots = \sum_{n=0}^{\infty} \frac{1}{(2^{p-1})^n}$$

$$= \frac{1}{1 - \frac{1}{2^{p-1}}} = \frac{2^{p-1}}{2^{p-1} - 1}$$

Now since $\frac{1}{2^{p-1}} < 1 \because p > 1 \therefore \sum \frac{1}{2^{p-1}}$ is cgt & hence by the comparison test $\sum \frac{1}{n^p}$ is cgt

case(2): $p=1$

$$\sum \frac{1}{n^p} = \sum \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

$$= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \dots$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \dots$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \text{ which is dgt } \infty$$

$\sum \frac{1}{n}$ is dgt by comparison test

case(3): $p < 1$

$$\sum \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots$$

$$> 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

$$> \sum \frac{1}{n} \text{ which is dgt}$$

$\therefore \sum \frac{1}{n^p}$ is dgt for $p < 1$.

Since $2^p < 2 \because p < 1$
 $\frac{1}{2^p} > \frac{1}{2}$

$$p < 1 \\ n^p < n \\ \frac{1}{n^p} > \frac{1}{n}$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \sum \frac{1}{n^2}$$

Now for $n > 2$

$$n^n > 2^n$$

$$\Rightarrow \frac{1}{n^n} < \frac{1}{2^n} \quad \text{Consider } \sum u_n = \sum \frac{1}{n^n} \quad \& \quad \sum v_n = \sum \frac{1}{2^n}$$

$$\frac{1}{n^n} < \frac{1}{2^n} \quad \text{i.e. } u_n < v_n \quad \forall n > 2$$

$\sum v_n = \sum \frac{1}{2^n}$ is cgt being a geometric series with $\text{c.r. } \frac{1}{2} < 1$
 $\therefore$ by comparison test $\sum \frac{1}{n^n}$ is cgt.

$$\text{eg: } \frac{1}{\log 2} + \frac{1}{\log 3} + \frac{1}{\log 4} + \dots + \frac{1}{\log n} + \dots$$

$$\text{sol: } \log n < n \quad \forall n > 1$$

$$\therefore \frac{1}{\log n} > \frac{1}{n}$$

$$\sum u_n = \sum \frac{1}{\log n} \quad \& \quad \sum v_n = \sum \frac{1}{n}$$

$$\& \quad u_n > v_n \quad \forall n > 1$$

$\sum v_n = \sum \frac{1}{n}$ is dgt by harmonic series. ($p \leq 1$ case)

$\therefore \sum \frac{1}{\log n}$ is dgt.

$$\text{eg: } \frac{1}{4 \cdot 6} + \frac{\sqrt{3}}{6 \cdot 8} + \frac{\sqrt{5}}{8 \cdot 10} + \dots = \sum u_n$$

$$u_n = \frac{\sqrt{2n-1}}{(2n+2)(2n+4)}$$

To find v_n we approximate $u_n \sim \frac{\sqrt{2n}}{(2n)(2n)} \sim \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$

$$\sum v_n = \sum \frac{1}{n^{3/2}}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{u_n}{v_n} &= \lim_{n \rightarrow \infty} \frac{\sqrt{2n-1}}{(2n+2)(2n+4)} \times n^{3/2} = \lim_{n \rightarrow \infty} \sqrt{\frac{2n-1}{n}} \left(\frac{n}{2n+2} \right) \left(\frac{n}{2n+4} \right) \\ &= \lim_{n \rightarrow \infty} \sqrt{2 - 1/n} \left(\frac{1}{2 + 2/n} \right) \left(\frac{1}{2 + 4/n} \right) = \sqrt{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2\sqrt{2}} \end{aligned}$$

$\therefore \lim_{n \rightarrow \infty} \frac{u_n}{v_n}$ is non zero & finite

$\therefore \sum u_n$ & $\sum v_n$ cgt or dgs together

Since $\sum v_n = \sum \frac{1}{n^{3/2}}$ is cgt $\therefore \sum u_n$ is cgt.

eg: $\frac{1}{1^2+x^2} + \frac{1}{2^2+x^2} + \frac{1}{3^2+x^2} + \dots$

$$u_n = \frac{1}{n^2+x^2} \quad v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+x^2} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{x^2}{n^2}} = 1 \neq 0 \text{ \& finite}$$

$\therefore \sum u_n$ & $\sum v_n$ cgs or dgs together

$\sum v_n$ cgs $\therefore \sum u_n$ cgs.

D'Alembert Ratio test - If $\sum u_n$ is +ve term series such that

$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = k$ then $\sum u_n$ cgs if $k < 1$ & dgs if $k > 1$. Test

fails for $k=1$.

~~Hint~~ Hint for the proof of test (cases $k > 1$ & $k < 1$)

For $k=1$ $\sum u_n = \sum \frac{1}{n}$ which is dgt series.

$$u_n = \frac{1}{n} \quad u_{n+1} = \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{1}{n+1} \cdot n = 1.$$

$\sum u_n = \sum \frac{1}{n}$ which is dgt series

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{1}{(n+1)^2} \cdot n^2 = 1.$$

$\therefore$ test can't be used for $k=1$

eg: $\frac{1}{1+x} + \frac{1}{1+2x^2} + \frac{1}{1+3x^3} + \dots \quad (x > 1)$

$$U_n = \frac{1}{1+nx^n} \quad ; \quad U_{n+1} = \frac{1}{1+(n+1)x^{n+1}}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} &= \lim_{n \rightarrow \infty} \frac{1+nx^n}{1+(n+1)x^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + x^n}{\frac{1}{n} + \left(\frac{n+1}{n}\right)x^{n+1}} = \frac{x^n}{x^{n+1}} = \frac{1}{x} \end{aligned}$$

But $x > 1 \quad \therefore \frac{1}{x} < 1$.

$\therefore \lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} < 1 \quad \therefore$ cgt series.

eg: $1 + \frac{2^2}{2!} + \frac{3^2}{3!} + \frac{4^2}{4!} + \dots$

$$U_n = \frac{n^2}{n!} \quad U_{n+1} = \frac{(n+1)^2}{(n+1)!}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} &= \frac{(n+1)^2}{(n+1)!} \cdot \frac{n!}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{(n+1)} \cdot \left(\frac{n+1}{n}\right)^2 \\ &= \lim_{n \rightarrow \infty} (n+1) \cdot \frac{1}{n^2} \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{1}{n^2}\right) = 0 < 1 \quad \therefore \text{cgt series.} \end{aligned}$$

Cauchy's Root test —

Let $\sum U_n$ is a +ve term series & $\lim_{n \rightarrow \infty} (U_n)^{1/n} = l$

If $l < 1$ then $\sum U_n$ cgs & if $l > 1$ then $\sum U_n$ dgs

Hint for the proof of test.

Test fails for $l=1$: e.g. $\sum \frac{1}{n}$ is dgt & $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right)^{1/n} = 1$

$\sum \frac{1}{n^2}$ is cgt & $\lim_{n \rightarrow \infty} \left(\frac{1}{n^2}\right)^{1/n} = 1$.

$$\text{eg: } 2 + \left(\frac{3}{2}\right)^4 + \left(\frac{4}{3}\right)^9 + \left(\frac{5}{4}\right)^{16} + \dots$$

$$u_n = \left(\frac{n+1}{n}\right)^{n^2}$$

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\left(\frac{n+1}{n}\right)^{n^2} \right]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e > 1 \text{ dgt.}$$

$$\text{eg: } \frac{1}{3}x + \frac{1 \cdot 2}{3 \cdot 5}x^2 + \frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7}x^3 + \dots \quad (x > 0)$$

$$u_n = \frac{1 \cdot 2 \cdot \dots \cdot n}{3 \cdot 5 \cdot \dots (2n+1)} x^n \quad (\text{apply Raabe's test}).$$

$$\text{eg: } \sum_{n=1}^{\infty} n! e^{-n^2}$$

$$u_n = n! e^{-n^2}$$

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} (n! e^{-n^2})^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \boxed{n!}^{\frac{1}{n}} \cdot e^{-\frac{n^2}{n}} = \lim_{n \rightarrow \infty} \boxed{n!}^{\frac{1}{n}} \cdot e^{-n}$$

$$= \lim_{n \rightarrow \infty} \boxed{n!}^{\frac{1}{n}} \cdot \lim_{n \rightarrow \infty} e^{-n} = 1 \cdot 1 = 1 \quad \therefore \text{test fails.}$$

Now we use comparison test

$$e^n > n \quad \forall n > 0$$

$$e^{n^2} > n^2 \quad \forall n.$$

$$\frac{1}{e^{n^2}} < \frac{1}{n^2}$$

$$n! e^{-n^2} < \frac{n!}{n^2} = \frac{1}{n^2} = u_n$$

$$\therefore \sum u_n = \sum \frac{1}{n^2} \text{ is cgt } \therefore \sum e^{-n^2} \text{ is cgt.}$$

$$\text{eg: } \sum_{n=1}^{\infty} (n^k - 1)^n$$

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} [(n^k - 1)^n]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} n^k - 1$$

$$= 1 - 1 = 0 < 1$$

$\therefore$ by root test, series cgs